RW–11057

Sub. Code
521303

M.Sc. DEGREE EXAMINATION, APRIL 2016

Physics

MICROPROCESSOR AND ELECTRONIC INSTRUMENTATION

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. List out the logical operations of ALU.

2. What is a timing unit?

3. Draw a flow chart for conditional structure.

4. Give any five branch group instruction sets.

5. What are advantages for 16 bit microcontroller?

6. How many bytes are available in internal RAM in 8051?

7. What is a tri-state buffer?

8. What is memory mapped I/O technique?

9. Define photoconductivity.

10. Define piezoelectric effect.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain about the accumulator and stack pointer.

Or

(b) Discuss about the direct addressing and register addressing modes.

12. (a) Explain the instruction set LXI, LDAX, XCHG, DAA and CMC.

Or

(b) Describe CALL and RET using subroutine.

13. (a) Write a program for the multiplication of two 8 bit numbers.

Or

(b) Discuss about the operational features of memory interfacing of 8051.

14. (a) Describe the DMA controller 8257.

Or

(b) Write a programme for a stepper motor for 8085.

15. (a) Explain about the resistor ladder network (R-2R) method.

Or

(b) Discuss about the successive approximation method.

Part C $(3 \times 10 = 30)$

Answer any **three** of the following.

16. Draw a block diagram of 8085 and explain each block.

17. Write a program to find the largest number in a data array.

18. Discuss the architecture of 8051.

19. Explain the communication interface of 8251 with a help of block diagram.

20. Describe in detail the classification of transducers.

RW–10845

Sub. Code
521202

M.Sc. DEGREE EXAMINATION, APRIL 2016

Second Semester

Physics

MATHEMATICAL PHYSICS – II

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** the questions.

All questions carry equal marks.

1. Define the essential singularity.

2. Write the equation of Laurent series.

3. What is the difference between 1st and 2nd order partial differential equations?

4. State the properties of Green's function.

5. State the Sturm-Liouville theory.

6. Define the degeneracy.

7. What are contravariant and covariant vectors?

8. Define the fully anti symmetric tensor of rank N.

9. What is the difference between reducible and irreducible representations?

10. Mention the crystal symmetry operators.

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) Write a short note on Schwartz reflection principle.

Or

(b) Explain the residue theorem.

12. (a) Write a detailed note on wave equation.

Or

(b) Derive Green's function.

13. (a) Obtain Self-Adjoint ODEs.

Or

(b) Explain the Hermitian operators.

14. (a) Discuss about the isotropic tensor.

Or

(b) Derive the equations for Lorentz transformation of coordinates.

15. (a) State and prove the orthogonality theorem.

Or

(b) Explain representation theory in quantum mechanics.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Explain in detail about mapping and conformal mapping.

17. Derive the Laplace and Poisson equations.

18. Explain in detail about the completeness of Eigen functions.

19. Discuss the outer and inner product of tensors and derive the equations for them.

20. Explain the character of a representation, construction of character tables and decomposition of reducible representations.

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RW–10846

Sub. Code
521203

M.Sc. DEGREE EXAMINATION, APRIL 2016

Second Semester

Physics

ELECTROMAGNETIC THEORY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

All questions carry equal marks.

1. Under what condition will the force on a current carrying wire placed in a magnetic induction be zero?

2. Can there be vector potential with zero magnetic field?

3. Define Snell's law.

4. What is Brewster angle?

5. What is the difference between dispersion and scattering?

6. Define polarization.

7. Distinguish between TE and TM modes.

8. Define the characteristic impedance and the propagation constant.

9. What is plasma?

10. What is pinch effect?

Section B (5 × 5 = 25)

Answer **all** the questions choosing either (a) or (b).

11. (a) Derive the wave equation in terms of scalar potential.

Or

(b) Find the electric field intensity and the electric potential due to a dipole.

12. (a) Write the boundary conditions for electrodynamics in the case of linear media.

Or

(b) Explain polarization and propagation vectors.

13. (a) Distinguish the normal and anomalous dispersions.

Or

(b) Derive Lorentz-Lorentz formula.

14. (a) Explain the resonant cavities in TM mode.

Or

(b) Discuss the working principle of a klystron.

15. (a) Explain the motion of a charged particle in a uniform magnetic field.

Or

(b) Discuss the general properties of plasma.

Section C

$(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. State and prove first and second uniqueness theorem.

17. Derive the reflection and transmission coefficients at normal incidence on the boundary between two linear media.

18. Derive Maxwell's equation in matter and also write down the constitutive relations among fields.

19. Explain the construction and working of a Gunn diode.

20. Derive the wave equations for E and B field in vacuum. Discuss its energy and momentum.

RW–10847

Sub. Code
521506

M.Sc. DEGREE EXAMINATION, APRIL 2016

Second Semester

Physics

DIGITAL ELECTRONICS PRINCIPLES

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. What is hexadecimal odometer?

2. Illustrate with example 1's complement method of subtraction.

3. Give the truth table for Exclusive OR gate. Write down the uses of this gate.

4. Explain 'Rolling' and 'Overlapping ' techniques in a Karnaugh map.

5. What is an encoder? When it is used?

6. Explain the use of binary multiplier.

7. Convert an R-S flip flop into a clocked R-S flip flop.

8. When you will call a counter as asynchronous counter?

9. Mention the function of programmable counter

10. What are demultiplexers?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

All questions carry equal marks.

11. (a) What is excess 3 code? Explain the method of converting decimal number into its excess-3 code with an example. How it is used with BCD numbers?

Or

(b) Explain in detail parity method for error detection and corrections with examples.

12. (a) State and prove DeMorgan's theorems.

Or

(b) Explain Quine-McClausky method of minimization procedure with example.

13. (a) Describe the working of a binary divider with necessary diagram.

Or

(b) Explain the 2's complement method of subtraction with two examples.

14. (a) Show how to convert a J-k flip flop into a D-type flip flop? Give its truth table.

Or

(b) Explain the principle of ring counter and discuss its uses.

15. (a) Draw the diagram for a mod-10 counter and explain its operation.

Or

(b) Construct a 3 bit synchronous counter using J-K flip flop.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Explain the method of converting the binary numbers to octal numbers. Convert the following three binary numbers to octal numbers

(a) 10101111

(b) 1101.0110111

(c) 1010011.101101.

17. Minimize the function $F = \sum m(0,1,2,3,11,12,14,15)$ using Karnaugh map, and realize in NOR logic.

18. Draw the full adder circuits using (a) NAND gates only and (b) NOR gates only.

Explain the functioning of the circuits and show that the output is that of a full adder.

19. Explain the action of master slave flip flop with a neat diagram. How the race around condition is eliminated in this system.

20. Give the circuits of direct coupled edge triggered D flip flop and explain its operation with necessary diagrams.

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RW–10848

Sub. Code
521401

M.Sc. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

Physics

CONDENSED MATTER PHYSICS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define the primitive cell.

2. Draw a diagram for FCC lattice.

3. What are phonons?

4. Define Fermi energy.

5. What is a dielectric material?

6. List the different polarizabilities.

7. Classify the magnetic materials.

8. Illustrate a spin wave with a diagram.

9. Define superconductivity.

10. Mention the formation of Cooper pairs.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the symmetry operations in crystals.

Or

(b) Find the Miller indices of a plane whose intercepts along the X,Y and Z axes are 2,4 and 6.

12. (a) Derive the dispersion relation for a monoatomic lattice.

Or

(b) Explain the Hall effect with a suitable diagram.

13. (a) Discuss about the local electric field present in a dielectric.

Or

(b) Explain the piezoelectricity with an example.

14. (a) Discuss the Langevin's theory of paramagnetism.

Or

(b) How are the hard and soft magnetic materials classified?

15. (a) Distinguish between the type I and type II superconductors.

Or

(b) Write a note on high temperature superconductors.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Draw the fourteen Bravais lattices in three dimension.

17. Explain Kronig-Penny model on the basis of the band theory of solids.

18. Describe the types of polarizabilities present in a dielectric.

19. Derive the dispersion relation for ferromagnetic magnons.

20. Discuss the D.C. Josephson tunneling effect.

RW–10849

Sub. Code
521402

M.Sc. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

Physics

NUCLEAR AND PARTICLE PHYSICS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

All questions carry equal marks.

1. What are mesons?

2. What is internal conversion?

3. Write out the shell configuration for ${}_{30}ZN^{67}$.

4. State the limitations of the shell model of nucleus.

5. Differentiate endoergic and exoergic reactions.

6. Complete the following reaction

   ${}^{32}S + \alpha \rightarrow$ ———————— $+ \gamma$

7. What is called differential cross section?

8. Why the study of p-p scattering is capable of much higher accuracy then n-p scattering?

9. Write down Gellmann-Okubo mass formula.

10. Define strangeness quantum numbers.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) Define parity. Explain the violation of parity conservation during β-decay process.

Or

(b) Write a note on neutrino helicity.

12. (a) Determine the ground state spin and parity of the following nucleus :

(i) ${}_{8}O^{17}$

(ii) ${}_{5}B^{11}$.

Or

(b) Write short notes on magnetic moments and Schmidt lines.

13. (a) State and explain the assumptions of the compound nucleus hypothesis.

Or

(b) Explain the ground state properties of deuteron.

14. (a) Give the theory of resonances in scattering and reaction cross-sections in nuclear reactions.

Or

(b) Classify and explain the types of nuclear reactors.

15. (a) Give an account of quark structure of mesons and baryons.

Or

(b) Which of the following reactions are allowed and forbidden under the conservation of strangeness, conservation of baryon number and conservation of charge?

(i) $\pi^+ + n \rightarrow \Lambda^0 + K^+$

(ii) $\pi^- + p \rightarrow K^- + \Sigma^+$.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Describe Fermis theory of beta decay and determine the mass of the neutrino by Fermi-Kurie plot.

17. Describe Bohr-Wheeler's theory on nuclear fission. What is the fissionability parameter and obtain the condition for spontaneously fission?

18. Explain in detail the effective range theory of n-p scattering at low energies.

19. Derive the four factor formula for a thermal nuclear reactor.

20. Explain the baryon octet and baryon decuplet symmetry scheme of elementary particle.

RW–10850

Sub. Code
521403

M.Sc. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

Physics

MATERIALS SCIENCE

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define viscoelasticity.

2. How do the metals get corroded?

3. State the law of diffusion.

4. What is meant by lattice misfit?

5. Draw the energy level diagram of He-Ne laser.

6. The indirect band semiconductors do not lase. Why?

7. Name a few carbon matrix composites.

8. Define the polymer-matrix composite.

9. What are glassy materials?

10. Define the acoustic optical effect.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the process of addition polymerization.

Or

(b) Discuss about the corrosion resistive materials.

12. (a) How are the thin films fabricated by thermal evaporation method?

Or

(b) Discuss about the preparation of thin films by liquid phase epitaxy method.

13. (a) How does the lasing action take place in a dye laser?

Or

(b) Explain the lasing action in He-Cd laser.

14. (a) Discuss about the cement-matrix composite.

Or

(b) Explain the ceramic-matrix composite.

15. (a) Discuss about the novel properties of shape memory alloys.

Or

(b) Explain the preparation of Nitinol alloys.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss about the relationship between the structure and the properties of polymers.

17. Explain the applications of Compound semiconductors

18. Describe the lasing action in CO_2 laser.

19. Discuss in detail about the applications of composites.

20. Discuss the applications of MEMS to micro-actuators and micro accelerometers.

RW–10851

Sub. Code
521510

M.Sc. DEGREE EXAMINATION, APRIL 2016

Fourth Semester

Physics

CRYSTAL GROWTH

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Define space lattice.

2. What are primary and secondary nucleations?

3. Define solubility.

4. What is meant by syneresis?

5. What is meant by zone refining?

6. What are the commonly used solvents in flux growth?

7. What is chemical vapour deposition?

8. Define Miller indices.

9. Classify the types of defects in crystals.

10. State the rule for the conventional orientation of crystals.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Discuss the crystal symmetry operations with necessary diagrams.

Or

(b) Explain the theory of surface energy.

12. (a) Explain Meir's solubility diagram.

Or

(b) Elucidate the mechanism of gel formation in silica gel media.

13. (a) Discuss the method of crystal growth using straight tube method.

Or

(b) List the merits and demerits of Bridgeman method.

14. (a) Describe briefly the liquid encapsulation in crystal growth process.

Or

(b) Describe the method of zone refining process and explain how is purification achieved.

15. (a) Briefly explain the orientation of crystals by X-ray method.

Or

(b) Write a short note on etching.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Explain in detail the two dimensional nucleation theories.

17. Explain the growth of single crystals by temperature gradient method.

18. Narrate the diagrams of apparatus used by Czochralski and explain the attainment of equilibrium thermodynamics.

19. Describe the method of growing single crystals by chemical vapour deposition technique. What is the uniqueness in this method?

20. Give details of the qualitative study on the properties of crystals.

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RW–10942

Sub. Code
521101

M.Sc. DEGREE EXAMINATION, APRIL 2016

Physics

CLASSICAL MECHANICS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. State the D'Alembert's principle.

2. What are Ignorable coordinates?

3. Show that the transformation $Q = 1/p$, $P = qp^2$ is canonical.

4. State the Hamilton-Jacobi theory.

5. Give the theorems about the moment of inertia.

6. Write Euler's equations of motion for a rotating rigid body.

7. What do you mean by relativity? State the two postulates of relativity.

8. Rest mass of an electron is 9.1×10^{-31} kg. Calculate its mass when it is moving with a velocity 0.8 c.

9. What is single stage and multi stage Rocket?

10. Define specific impulse.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) State and prove the principle of virtual work.

Or

(b) A particle of mass m attached to a spring of force constant k is executing a SHM about the equilibrium position. At an instant t, the displacement of the particle from the equilibrium position is x and its velocity is v. Write down Lagrangian of the system and find the equation of motion.

12. (a) Discuss about the Hamilton-Jacobi theory.

Or

(b) Show that the transformation $P = \frac{1}{2}(p^2 + q^2)$, $Q = \tan^{-1}\frac{q}{p}$ is canonical.

13. (a) Explain the principle axes inertia.

Or

(b) Derive an expression for Kinetic energy of a rigid body.

14. (a) Discuss the length contraction under Lorentz transformation.

Or

(b) Calculate the velocity at which the mass of a particle becomes double of its rest-mass.

15. (a) Write any two exhaust speed parameters.

Or

(b) Derive an equation for the thrust of a rocket.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Derive the Lagrangian equation of motion under conservative force field. Construct Lagrangian equation of motion for a simple pendulum.

17. State and prove principle of least action.

18. Derive the angular velocity and kinetic energy of a rotating body in terms of Euler angle.

19. Discuss in detail the Lorentz transformations

20. Describe in detail the optimization of a multi-stage rocket.

RW–10943

Sub. Code
521102

M.Sc. DEGREE EXAMINATION, APRIL 2016

Physics

MATHEMATICAL PHYSICS – I

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Obtain the value of $a.b$ if a and b are perpendicular to each other.

2. Evaluate $div\mathbf{r}$ if $\mathbf{r} = \mathbf{i}x + \mathbf{j}y + \mathbf{k}z$, ***i, j, k*** being the unit vectors along *X, Y, Z* axes, respectively.

3. When is a matrix said to be idempotent?

4. Define the rank of a matrix,

5. Show that $\beta(m,n) = \beta(n,m)$.

6. For Bessel functions, what is $J_{1/2}(x)$ equal to?

7. If $g(\omega)$ is the Fourier transform of $f(t)$, then show that the Fourier transform of the first derivative (df/dt) is given by $i\omega g(\omega)$.

8. If $f(s)$ is the Laplace transform of $F(t)$, then show that the Laplace transform of $\exp(at).F(t) = f(s-a)$.

9. What is the probability that a leap year selected at random contains 53 Sundays?

10. Six cards are drawn at random from a pack of 52 cards which contains equal number of red and black cards. What is the probability that 3 will be red and three will be black?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Prove that $div\,curl A = 0$.

Or

(b) Deduce the Laplacian in cylindrical coordinates.

12. (a) Prove that every square matrix can be uniquely expressed as the sum of a symmetric and a skew-symmetric matrix.

Or

(b) Show that the matrix $\begin{bmatrix} 1/\sqrt{2} & i/\sqrt{2} \\ -i/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$ is unitary.

13. (a) Prove that $\Gamma(m)\Gamma(l-m) = \pi / \sin(m\pi)$.

Or

(b) For Legendre polynomials, show that $P_n(-x) = (-1)^n P_n(x)$.

14. (a) Find the finite cosine Fourier transform of $\sin(ax)$ in the interval $(0, \pi)$.

Or

(b) Find the Laplace transform $L\{t^n\}, n > -1$. Hence find it for $n = 1/2$.

15. (a) Obtain the expression for the mean of binomial distribution.

Or

(b) Derive the second moment of Poisson distribution.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Evaluate $\iint (\nabla \times F).nds$ for the vector $\mathbf{F} = (x^2 + y - 4)i + 3xyj + (2xz + z^2)k$, over the surface of hemisphere $(x^2 + y^2 + z^2) = 16$ lying above $x - y$ plane, n being the unit normal vector.

17. Diagonalize the matrix $\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

18. State and prove the orthogonality relation for Hermite polynomials.

19. State and prove the convolution theorem of Laplace transforms.

20. Show that the mean deviation from the mean of the normal distribution is about $(4/5)^{th}$ of its standard deviation.

RW–10944

Sub. Code
521501

M.Sc. DEGREE EXAMINATION, APRIL 2016.

Physics

ELEMENTARY NUMERICAL ANALYSIS

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

All questions carry equal marks.

1. What is called round off error? Give an example.

2. What are the demerits of bisection method?

3. What is called linear interpolation?

4. For a set of four data, what is the maximum order of divided differences needed for interpolation?

5. What are the uses of forward interpolation and backward interpolation formulas?

6. Write the algorithm of trapezoidal rule.

7. What is the truncation error of Simpson's one-third rule?

8. Write down the algorithm of Euler's method for the solution of first-order ODE.

9. What is called a singular matrix?

10. What is called partial pivoting?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

All questions carry equal marks.

11. (a) Explain the procedure of least-squares fitting of a data set with an exponential function.

Or

(b) Estimate the order of convergence in Newton-Raphson method.

12. (a) Discuss the computational procedure of non-linear interpolation.

Or

(b) Deduce Newton's forward interpolation formula.

13. (a) Evaluate the integral $\int [1/(1+x)]dx$ between the limits (0, 6), by using the trapezoidal rule with a step interval of $h = 1$.

Or

(b) Derive Simpson's one-third rule.

14. (a) Discuss Taylor's series method of solving first-order ODE and write down the corresponding algorithm.

Or

(b) Find the solution $y(x)$ at $x = 0.2$ of the differential equation $(dy/dx) = 1 - 2xy$, given that $y(0) = 0$, by taking the step interval $h = 0.1$.

15. (a) Discuss the computational procedure of Gaussian elimination to solve a system of linear equations.

Or

(b) Discuss the LU-factorization procedure of solving simultaneous equations.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Compute the root of $x^3 - 9x + 1 = 0$ that lies between 2 and 4 by using secant method.

17. From the following data set, find the value of y at $x = 84$.

x :	40	50	60	70	80	90
y :	184	204	226	250	276	304

18. Estimate the value of $\log_e\left(2^{1/3}\right)$ by evaluating $\int \left[x^2/\left(1+x^3\right)\right] dx$, between the limits (0, 1), using Simpson's one-third rule with a step size of $h = 0.25$.

19. Applying RK-4 method, obtain the values of $y(x)$ at $x = 0.1$ and 0.2, for the differential equation $(dy/dx) = -y$, given that $y(0) = 1$. Take $h = 0.1$.

20. Solve by Gauss-Siedel method :

$8x - 3y + 2z = 20$, $4x + 11y - z = 33$ and $6x + 3y + 12z = 35$.

RW–10945

Sub. Code
521301

M.Sc. DEGREE EXAMINATION, APRIL 2016

Physics

CHEMICAL PHYSICS – I

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are bonding and anti-bonding orbitals?

2. How does sp^2 hybridization occur?

3. Distinguish between rigid and non-rigid rotators.

4. What is called hyperfine interaction?

5. What is known as harmonic approximation of vibratinoal motion?

6. What are the selection rules for the transitions between vibrational states?

7. What is the called the inverse Raman effect?

8. Mention the effect of rotational motion on the vibrational energy spectrum of a diatomic molecule.

9. What is called the nuclear spin resonance?

10. Mention the applications of ESR.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the valence bond theory of molecules.

Or

(b) Explain the effect of hybridization of orbitals on the electronic energy spectrum of molecules.

12. (a) Discuss the general theory of a non-rigid rotator and obtain its rotational energy spectrum.

Or

(b) Explain the influence of a static electric field on the rotational spectrum of a molecule.

13. (a) Explain the general theory of IR spectra of molecules.

Or

(b) Discuss the quantum theory of Franck-Condon principle.

14. (a) Explain the experimental method of studying the anti-stokes Raman scattering by molecules and mention its merits and demerits.

Or

(b) Discuss the theory of multi-photon absorption by molecules.

15. (a) Outline the general theory of spin-lattice relaxation process.

Or

(b) Discuss the fundamental theory of NQR spectroscopy.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Using the molecular orbital theory, explain the theory of obtaining the bonding and anti-bonding energies of a H_2 molecule.

17. Discuss the theory of rotational Raman spectra of a diatomic molecule.

18. Discuss in detail the method of studying the molecular structure using Raman spectroscopy.

19. Discuss the theory and experimental procedure of studying the photo-acoustic Raman scattering by molecules.

20. Discuss in detail the application of Mossbauer spectroscopy to study the molecular structure.

RW–10946

Sub. Code
521302

M.Sc. DEGREE EXAMINATION, APRIL 2016

Physics

QUANTUM MECHANICS – II

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define angular momentum and write its uncertainty relation.

2. Write down the Pauli spin matrices.

3. Are the wave functions of bosons symmetric? Why?

4. What are the uses of Dirac's bra and ket vectors?

5. What is self-consistent field approximation?

6. What is the advantage of Hartree-Fock method over Hartree method?

7. What is the relativistic equation used for treating spin-1/2 particles?

8. What is known as antimatter?

9. Define the particle density operator in Fock space.

10. Write the relation between Hamiltonian density and total Hamiltonian.

Part B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) Evaluate the commutator $\left[L^2, L\right]$ L being the angular momentum operator.

Or

(b) Derive Pauli's spin matrices.

12. (a) Show that the permutation operator is hermitian.

Or

(b) Discuss the interaction picture of representing dynamical variables.

13. (a) Explain the Hartree method of solving a many-body problem.

Or

(b) Explain the use of quantum mechanical electronic structure of atoms in the design of modern periodic table of elements.

14. (a) Derive Dirac Hamiltonian.

Or

(b) For the Dirac Γ-matrices, show that $Tr(\Gamma i) = 0$ for $\Gamma i \neq I$.

15. (a) Derive an expression for the canonical momentum by using the Lagrangian density.

Or

(b) Explain the ground state properties of a field with N spinless fermions.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the theory of addition of two angular momenta and obtain the CG coefficients for $j = 1/2$ system.

17. Solve the harmonic oscillator by using the matrix representation method.

18. Explain in detail the Thomas-Fermi model of atoms.

19. Solve the Klein-Gordon equation for free particles and explain the important characteristics of the energy spectrum.

20. Discuss the theory of quantizing the normal modes of EM field.

RW–10844

Sub. Code
521201

M.Sc. DEGREE EXAMINATION, APRIL 2016

Second Semester

Physics

QUANTUM MECHANICS — I

(2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(10 \times 2 = 20)$

Answer **all** questions.

All questions carry equal marks.

1. What is known as wave-particle duality?

2. Can we write the uncertainty relation with two position variables? Justify.

3. What is called a central potential?

4. What are spherical harmonics?

5. What are the limitations of perturbation theory?

6. When does WKB method fail?

7. What are the conditions for stimulated emission to occur?

8. Mention the semiclassical limit of radiation theory.

9. What is called scattering amplitude?

10. When does Born approximation fail?

Section B (5 × 5 = 25)

Answer **all** the questions.

All questions carry equal marks.

11. (a) Derive the time-dependent Schroedinger equation.

Or

(b) Verify whether the eigen-functions $\psi_1(X) = A\cos kx$, and $\psi_2(X) = B\sin kx$, of a particle in a box, are orthogonal.

12. (a) Treating as a rigid rotator, obtain the rotational energy spectrum of a hydrogen molecule.

Or

(b) Derive the radial Schroedinger equation.

13. (a) Discuss the non-degenerate first-order perturbation theory.

Or

(b) Apply the WKB approximation to a one-dimensional harmonic oscillator and obtain the energy spectrum.

14. (a) Discuss Einstein's theory of radiation for a two-level system and mention the significance of the results.

Or

(b) Briefly discuss the quantum theory of radiation within the frame-work of time-dependent perturbation theory.

15. (a) Discuss the theory of scattering using the Born approximation.

Or

(b) Discuss briefly the theory of scattering by screened Coulomb potential.

Section C

$(3 \times 10 = 30)$

Answer any **three** questions.

All questions carry equal marks.

16. Discuss the theory and results of a one-dimensional harmonic oscillator.

17. Deduce the bound state energy spectrum of hydrogen atom.

18. Applying variational method, deduce the energy spectrum of a deuteron.

19. Explain the phenomenon of Rayleigh scattering using time-dependent perturbation theory.

20. Discuss the general theory of partial wave analysis in scattering.